

Hypothesis Testing

1. **Null Hypothesis (H_0):** There is no significant difference or effect.
2. **Alternative Hypothesis (H_1):** There is a significant effect i.e the given statement can be false.
3. **Degrees of freedom:** [Degrees of freedom \(df\)](#) in statistics represent the number of values or quantities in the final calculation of a statistic that are free to vary. It is mainly defined as sample size-one ($n-1$).
4. **Level of Significance (α):** This is the threshold used to determine statistical significance. Common values are 0.05, 0.01, or 0.10.
5. **p-value:** The [p-value](#) probability of observing results if H_0 is true.
 - If $p \leq \alpha$: reject H_0
 - If $p > \alpha$: fail to reject H_0
6. **Type I Error and Type II Error**
 - **Type I Error** that occurs when the null hypothesis is true, but the statistical test incorrectly rejects it. It is often referred to as a "false positive" or "alpha error."
 - [Type II Error](#) that occurs when the null hypothesis is false, but the statistical test fails to reject it. It is often referred to as a "false negative."
7. **Confidence Intervals:** A [confidence interval](#) is a range of values that is used to estimate the true value of a population parameter with a certain level of confidence. It provides a measure of the uncertainty or margin of error associated with a sample statistic, such as the sample mean or proportion.

Statistical tests

Parametric test are statistical methods that make assumption that the data follows normal distribution.

<u>Z-test</u>	<u>t-test</u>	<u>F-test</u>
Tests if a sample mean differs from a known population mean.	Compares means when population standard deviation is unknown.	Compares variances of two or more groups.
Population standard deviation is known and sample size is large.	Small samples or unknown population standard deviation.	To test if group variances are significantly different.
One-Sample Test: $Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$ Two-Sample Test: $Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	One- sample: $t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$ Two-Sample Test: $t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ Paired t-Test: $t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$ d= difference	$F = \frac{s_1^2}{s_2^2}$